Table 1: Experimental design.

Component	Specification
Task	Flight recommendation (Qiu et al., 2026); 3 flights per menu
Attributes	Departure time, duration, number of stops, price (scaled to $[0, 1]$)
Principals	50 simulated users, $\theta_i \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}^4$, stratified by $\|\theta_i\|$
Interaction	5 rounds; after each pick: "correct" or "incorrect, I prefer Flight k"
Evaluation	100 held-out menus at 6 checkpoints (after 0–5 rounds), no feedback
Decisions	$50 \times 605 = 30{,}250$ per agent
Outcomes	Accuracy; utility regret; share of achievable welfare
Inference	95% CIs from resampling users (10,000 draws)

Table 2: Agents compared.

Agent	Provider	Decoding	Role
Qwen 2.5 7B Instruct	Phala	$T = 0$	Replication target
Gemma 4 31B	Friendli	$T = 0$, no thinking	Current open model
Gemma 4 31B, thinking	Friendli	$T = 0$, 1,024-token cap[a]	Effect of reasoning
Bayesian Assistant	local	exact posterior	Information benchmark
Simple heuristics[b]	local	–	Pipeline checks

[a]Cut-off reasoning is followed by a forced answer; forced picks are flagged. [b]Random, always-first, copy-last-choice.

Table 3: Held-out accuracy (%) and utility regret by rounds of feedback. 95% CIs in brackets.

	Accuracy (%)			Regret
Agent	Round 0	Round 1	Round 5	Round 5
Bayesian Assistant	33.6	56.1	80.6 [77.6, 83.4]	0.043
Gemma 4 31B, thinking	37.6	47.9	62.7 [58.9, 66.3]	0.144
Gemma 4 31B	36.9	45.0	59.4 [55.5, 62.9]	0.164
Qwen 2.5 7B	36.6	37.0	38.0 [34.6, 41.8]	0.348
Random	33.8	33.2	33.3 [32.3, 34.3]	0.398

Paper, round 5: Bayesian Assistant ≈80%; Qwen 2.5 7B 37%; Gemma 2 9B 37%.

Table 4: Share of achievable welfare (%) by principal type, and the welfare-maximising number of examples n^* at cost c per example.

	Welfare (%)			n^* at $c =$		
Agent	Round 0	Round 5	Gain	1	2	4
Conventional principals ($n = 14$)						
Bayesian Assistant	50.2	91.4	+41.2	5	3	3
Gemma 4 31B, thinking	77.1	83.4	+6.3	5	0	0
Gemma 4 31B	73.5	80.3	+6.8	3	1	0
Qwen 2.5 7B	74.0	67.9	−6.1	0	0	0
Unusual principals ($n = 18$)						
Bayesian Assistant	48.8	89.6	+40.8	5	3	3
Gemma 4 31B, thinking	32.9	71.6	+38.7	3	3	3
Gemma 4 31B	32.8	68.3	+35.5	5	3	3
Qwen 2.5 7B	33.3	45.1	+11.8	2	2	1

Unusual: prefers pricier, longer, or more stops on ≥ 2 of 3 attributes. c in welfare points per example.

Table 5: Final-round accuracy (%): paper vs. this replication.

Agent	Qiu et al. (2026)	This study
Bayesian Assistant	≈ 80	80.6
Qwen 2.5 7B	37	38.0
Gemma 2 9B / Gemma 4 31B	37	59.4
Gemma 4 31B, thinking	–	62.7

$$U_i(f) = \theta_i^\top \phi(f) \tag{1}$$

$$R^A_{imt} = \max_{f \in M_m} U_i(f) - U_i\big(f^A_{imt}\big) \tag{2}$$

$$w^A_{imt} = \frac{U_i\big(f^A_{imt}\big) - \min_{f \in M_m} U_i(f)}{\max_{f \in M_m} U_i(f) - \min_{f \in M_m} U_i(f)} \tag{3}$$

$$W^A_i(t) = \frac{100}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} w^A_{imt}, \qquad \overline{W}^A_G(t) = \frac{1}{|G|} \sum_{i \in G} W^A_i(t) \tag{4}$$

$$NW^A_G(n; c) = \overline{W}^A_G(n) - c\, n, \qquad n^*(c) = \operatorname*{arg\,max}_{n \in \{0, \ldots, 5\}} NW^A_G(n; c) \tag{5}$$

$$D_i = \mathbf{1}\Big[\textstyle\sum_{j \in \{\text{dur, stops, price}\}} \mathbf{1}[\theta_{ij} > 0] \geq 2\Big] \tag{6}$$

where $\mathcal{M}_i$ is the set of held-out menus on which principal i is not indifferent between all three flights.